Azizul Zahid

Louisville, TN, USA | (+1) 8652327003
Email: azahid@vols.utk.edu
LinkedIn | GitHub | Website | Google Scholar

RSEARCH INTERESTS

LLM, Edge-AI, Reinforcement Learning, Health Robotics, Wearable Sensors

ACADEMIC CREDENTIALS

University of Tennessee, Knoxville **June'23 - Present**
PhD (ongoing) in Computer Engineering (3.97/4.00) *Knoxville, TN, USA*

University of Tennessee, Knoxville **June'23 - May'25**
M.Sc. in computer engineering (3.97/4.00) *Knoxville, TN, USA*

Bangladesh University of Engineering and Technology **Feb'17 - May'22**
B.Sc. in Electrical and Electronic Engineering (3.63/4.00) *Dhaka, Bangladesh*

RESEARCH EXPERIENCES

Graduate Research Assistant, EPIC LAB **June'23 - Present**
Supervisor: Dr. Sai Swaminathan
Electrical Engineering and Computer Science (EECS), University of Tennessee, Knoxville

WORK EXPERIENCES

Graduate Teaching Assistant Fall'23, Spring'24, Fall'24, Spring'25, Fall'25
Courses: Embedded Systems (ECE 455/555) Computer Organizations (COSC 230), Circuits I (ECE 201), Computer System Organization (ECE 356)
Electrical Engineering and Computer Science (EECS), University of Tennessee, Knoxville

Lecturer in Electrical and Computer Engineering October'22 - March'23
Southeast University, Dhaka, Bangladesh

PUBLICATIONS

Conferences

C1: "*PulseRide: A Robotic Wheelchair for Personalized Exertion Control with Human-in-the-Loop Reinforcement Learning*", **Azizul Zahid**, Bibek Poudel, Danny Scott, Jason Scott, Scott Crouter, Weizi Li, Sai Swaminathan, accepted in **IEEE/ACM CHASE 2025**.

W1: "*Toward Aligning Human and Robot Actions via Multi-Modal Demonstration Learning*", **Azizul Zahid**, Jie Fan, Farong Wang, Ashton Dy, Sai Swaminathan, Fei Liu, accepted in **ICRA 2025.**

Journals

J1: "*RadioGami: Battery-free, Micropower, Long-Range, Interactive Frequency Selective Surfaces*", Imran Fahad, Danny Scott, **Azizul Zahid**, Matthew Bringle, Srinayana Patil, Ella Bevins, Carmen Palileo, Sai Swaminathan, accepted in "*Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT) – 2025*".

J2: "*NeuroCamTags: Long-Range, Battery-free, Wireless Sensing with Neuromorphic Cameras*", Danny Scott, Matthew Bringle, Imran Fahad, Gaddiel Morales, **Azizul Zahid**, Sai Swaminathan, accepted in "*Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT) – 2024*".

J3: Mir Sayeed Muhammad, **Azizul Zahid**, Md. Asif Iqbal, "*BanglaNum - A Public Dataset for Bengali Digit Recognition from Speech*", arXiv preprint arXiv: 2403.13465, 2024

RELEVENT COURSEWORKS

Graduate level: Machine learning (COSC 522), Reinforcement Learning (ECE 517), Embedded Systems (ECE 555), Bio Inspired Computation (COSC 527), Deep Learning (COSC 525), Natural Language Processing (COSC 524), Human Computer Interaction (COSC 559), Computer Vision (ECE 574), Robot opt., est. & cont. (ECE 599)
Undergraduate level: Probability and Statistics, Linear Algebra, Calculus, Microprocessors and Embedded System, Digital Electronics, Solid State Devices, Communication I & II, Computer Programming theory & Lab, Digital Signal Processing, Control Systems

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, C/C++
Data mining: NumPy, Pandas
Data visualization: Matplotlib, Seaborn
Machine learning: Scikit-Learn
Deep learning: PyTorch
Others: Git, GitHub, Latex, MS office

PROJECTS

- "*FlowSync: Synergizing Heartbeat and Algorithms for Dynamic Fluid Control Enhancement.*" – Course project for ECE 517 (Fall'23)
- "*IoT-based Temperature Control System in Poultry Farm.*" – Course project for ECE 555 (Fall'23)
- "*Adaptive Neural Piloting: Exploring Neural Network Applications in Aviation Simulation.*" – Course project for COSC 525 (Spring'24)
- "*Gesture Recognition using Spiking Neural Network.*" – Course project for COSC 527 (Spring'24)

REFERENCES

Dr. Sai Swaminathan
Assistant Professor
EECS, UTK
Email: sai@utk.edu